

Mathematical and Logical Foundations of Computer Science – Logic

University of Birmingham

November 11, 2022

1 Propositional Logic

1.1 Syntax

The syntax of propositional logic formulas is defined by the following grammar, where a ranges over **atomic propositions** (atoms):

$$P ::= a \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \neg P$$

There are two special atoms: $\top$ which stands for True, and $\perp$ which stands for False.

We use the following precedence and associativity rules:

- **Precedence**: in decreasing order of precedence $\neg, \wedge, \vee, \rightarrow$.
- **Associativity**: all operators are right associative

1.2 Constructive Natural Deduction

$$\begin{array}{c}
 \frac{\perp}{A} [\perp E] \quad \frac{}{\top} [\top I] \quad \frac{\overline{A}^1 \quad \vdots \quad B}{A \rightarrow B}^1 [\rightarrow I] \quad \frac{A \rightarrow B \quad A}{B} [\rightarrow E] \\
 \\
 \frac{\overline{A}^1 \quad \vdots \quad \perp}{\neg A}^1 [\neg I] \quad \frac{\neg A \quad A}{\perp} [\neg E] \\
 \\
 \frac{A}{A \vee B} [\vee I_L] \quad \frac{A}{B \vee A} [\vee I_R] \quad \frac{A \vee B \quad A \rightarrow C \quad B \rightarrow C}{C} [\vee E] \\
 \\
 \frac{A \quad B}{A \wedge B} [\wedge I] \quad \frac{A \wedge B}{B} [\wedge E_R] \quad \frac{A \wedge B}{A} [\wedge E_L]
 \end{array}$$

1.3 Classical Natural Deduction

It includes all the Constructive Natural Deduction rules, plus:

$$\frac{}{A \vee \neg A} [LEM] \quad \frac{\neg \neg A}{A} [DNE]$$

1.4 Truth Tables

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

A	$\neg A$
T	F
F	T

$\top$
T

$\perp$
F

2 Predicate Logic

2.1 Syntax

The syntax of predicate logic is defined by the following grammar:

$$\begin{aligned} t &::= x \mid f(t, \dots, t) \\ P &::= p(t, \dots, t) \mid \neg P \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \forall x.P \mid \exists x.P \end{aligned}$$

There are two special predicate symbols of arity 0: $\top$ which stands for True, and $\perp$ which stands for False.

2.2 Natural Deduction

The Natural Deduction rules for Predicate Logic include all Proposition Logic rules plus the following rules:

$$\frac{P[x \backslash y]}{\forall x.P} [\forall I] \quad \frac{\forall x.P}{P[x \backslash t]} [\forall E] \quad \frac{P[x \backslash t]}{\exists x.P} [\exists I] \quad \frac{\exists x.P \quad \overline{P[x \backslash y]}^1}{Q} [\exists E]$$

Side conditions:

- for $[\forall I]$: y must not be free in any not-yet-discharged hypothesis or in $\forall x.P$
- for $[\forall E]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- for $[\exists I]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- for $[\exists E]$: y must not be free in Q or in not-yet-discharged hypotheses or in $\exists x.P$

2.3 Semantics

Given a signature: $\langle \langle f_1^{k_1}, \dots, f_n^{k_n} \rangle, \langle p_1^{j_1}, \dots, p_m^{j_m} \rangle \rangle$

- of function symbols f_i of arity k_i , for $1 \leq i \leq n$
- of predicate symbols p_i of arity j_i , for $1 \leq i \leq m$

a model M is a structure $\langle D, \langle \mathcal{F}_{f_1}, \dots, \mathcal{F}_{f_n} \rangle, \langle \mathcal{R}_{p_1}, \dots, \mathcal{R}_{p_m} \rangle \rangle$

1. of a non-empty domain D
2. interpretations $\mathcal{F}_{f_i} \in D^{k_i} \rightarrow D$ for function symbols f_i
3. interpretations $\mathcal{R}_{p_i} \subseteq D^{j_i}$ for function symbols p_i

A variable valuation v is a partial function from variables to D

Given a model M and a variable valuation v , we assign meaning to terms and formulas as follows:

- Meaning of terms:
 - $\llbracket x \rrbracket_v^M = v(x)$
 - $\llbracket f(t_1, \dots, t_n) \rrbracket_v^M = \mathcal{F}_f(\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle)$
- Meaning of formulas:
 - $\models_{M,v} \top$ is true
 - $\models_{M,v} \perp$ is false
 - $\models_{M,v} p(t_1, \dots, t_n)$ iff $\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle \in \mathcal{R}_p$
 - $\models_{M,v} \neg P$ iff $\not\models_{M,v} P$
 - $\models_{M,v} P \wedge Q$ iff $\models_{M,v} P$ and $\models_{M,v} Q$
 - $\models_{M,v} P \vee Q$ iff $\models_{M,v} P$ or $\models_{M,v} Q$
 - $\models_{M,v} P \rightarrow Q$ iff $\models_{M,v} Q$ whenever $\models_{M,v} P$
 - $\models_{M,v} \forall x.P$ iff for every $d \in D$ we have $\models_{M,(v,x \mapsto d)} P$
 - $\models_{M,v} \exists x.P$ iff there exists a $d \in D$ such that $\models_{M,(v,x \mapsto d)} P$